

Spectral Gaps of Hit-and-Run and Coordinate Hit-and-Run

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Abstract

For any convex body $\mathcal{K} \subset \mathbb{R}^n$ containing a unit ball, the spectral gap of Hit-and-Run is $\Omega(1/(n^2 C_{\text{PI}}))$, where C_{PI} is the Poincaré constant of the uniform distribution π over $\mathcal{K}$. This implies that Hit-and-Run converges to a distribution within χ^2 -divergence ε of the uniform distribution π in $O(n^2 C_{\text{PI}} \log(M/\varepsilon))$ steps from any starting distribution π_0 with $M = \chi^2(\pi_0 \parallel \pi)$, thus refining the known bound of $O(n^2 R^2 \log(M/\varepsilon))$ by Lovász and Vempala (2004) in terms of the outer radius R ; for nearly isotropic bodies, together with progress on the KLS conjecture, the complexity is $O(n^2 \log n \log(M/\varepsilon))$, improving the dimension dependence from cubic to nearly quadratic while maintaining logarithmic dependence on the initial distance. It was an open problem to connect the convergence of Hit-and-Run to Poincaré/KLS constants as was done for the Ball walk by Kannan, Lovász and Simonovits (1997). Unlike Hit-and-Run, the Ball walk has an unavoidable linear dependence on (a stronger notion) of the initial warmness.

We directly bound the spectral gap of the Hit-and-Run Markov chain by connecting it to functional isoperimetric constants, inspired by the recent analysis of In-and-Out. Rewriting the spectral gap first in terms of dual certificates leads to the Babuška–Aziz constant studied in the analysis of PDEs; it is asymptotically bounded by the Improved Poincaré constant of the target distribution, which we show can be bounded in terms of the usual Poincaré constant. The proof is based on duality and calculus, unlike known proofs of convergence for Hit-and-Run which are based on bounding the conductance. The same technique can be applied to Coordinate Hit-and-Run, resulting in a much improved mixing time of $O(n^3 C_{\text{PI}} \log(M/\varepsilon))$.

AI disclosure: A preliminary version of the main technique/proof/figure was suggested by GPT 5.6 Pro through guided prompting by the first author. The authors then verified, streamlined and wrote the full proof.

1 Introduction

Sampling a convex body is a fundamental algorithmic problem. Hit-and-Run, proposed independently by Boneh and Golan [BG79] and Smith [Smi84], is the following Markov chain (see Figure 1.1) when applied to a convex body $\mathcal{K} \subset \mathbb{R}^n$: at a current point $x \in \mathcal{K}$,

1. Sample a uniformly distributed random line ℓ through x .
2. Go to a uniform random point on the chord $\ell \cap \mathcal{K}$.

Hit-and-Run is an attractive, easy-to-implement process that has been widely used in practice. Coordinate Hit-and-Run (CHAR), also known as Gibbs sampling and introduced in 1971 [Tur71], is the variant that only uses axis-parallel chords and is also popular due to its low memory overhead.

Building on the seminal work of [Lov99], Lovász and Vempala [LV06a] showed that Hit-and-Run mixes from a cold start: in any convex body $\mathcal{K} \subset \mathbb{R}^n$ with $B^n \subseteq \mathcal{K} \subseteq RB^n$, it outputs a point within χ^2 -divergence ε of the uniform distribution over $\mathcal{K}$ in $O(n^2 R^2 \log(M/\varepsilon))$ steps from any starting distribution within χ^2 -divergence M of the target. This convergence rate is asymptotically optimal in terms of these parameters, as shown by a cylinder whose cross section is a unit ball and axis has length $2R$. It also means that Hit-and-Run mixes rapidly from any interior starting point (by applying the convergence bound to the distribution obtained after taking one step), the first random walk known to have this property for arbitrary convex bodies. More recently, CHAR was shown to have (higher) polynomial convergence rates [LV23, NS22, NRS25], with the last paper showing logarithmic dependence on the warm start parameter.

The study of efficiently sampling convex bodies has made much progress over the past few decades, starting from an initial polynomial-time algorithm with complexity n^{23} by Dyer, Frieze, and Kannan [DFK91]. The Ball walk introduced by Lovász [Lov90] has been particularly fruitful, leading to the current best complexity for sampling, general techniques for the analysis of Markov chains [LS90, LS93, KLS97, LV07], and novel classes of isoperimetric inequalities. One highlight is the Kannan–Lovász–Simonovits (KLS) conjecture [KLS95] which posits that halfspaces are asymptotically optimal isoperimetric cuts for convex bodies (and logconcave densities). The conjecture directly leads to a better bound on the convergence rate of the Ball walk, and has several other interesting mathematical consequences and connections. Another highlight is the localization method, first developed by Kannan, Lovász, and Simonovits [LS93, KLS95] to prove inequalities in high dimension and later generalized to a stochastic method by Eldan [Eld13]. For a more comprehensive account of these developments, we refer the reader to [Vem05, KV25a].

The Ball walk has a polynomial dependence on the warmness parameter¹, and this is unavoidable. To get around this, researchers have developed algorithmic solutions, notably annealing [LV06b, KV06, CV18], to arrange an $O(1)$ -warm start. From such a warm start, the Ball walk’s convergence rate is roughly $n^2 \|\text{cov}\|_{\text{op}} \psi_n^2$, where $\|\text{cov}\|_{\text{op}}$ is the operator norm of the covariance of the target distribution, and ψ_n is the KLS constant, which is known to be $O(\sqrt{\log n})$ and is conjectured to be $O(1)$. This is always a better bound than $n^2 R^2$ and is asymptotically better by a factor of n when the target is isotropic (i.e., $\|\text{cov}\|_{\text{op}} = 1$). Note that for convex bodies and logconcave densities, we have $C_{\text{PI}} \simeq \|\text{cov}\|_{\text{op}} \psi_n^2$, where the RHS is the squared reciprocal of the Cheeger constant.

On the other hand, despite its logarithmic dependence on the warmness parameter, there was no known connection between the convergence of Hit-and-Run and the KLS constant. Thus, a decade of progress on the latter, leading to improved complexities for sampling (near-)isotropic convex bodies, did not refine the bound for Hit-and-Run. Unlike the Ball walk, whose analysis directly uses Euclidean isoperimetry, the analysis of Hit-and-Run was based on an average isoperimetric inequality for the

¹In fact, the Ball walk uses a more stringent notion of warmness, namely $d\pi_0/d\pi \leq M$.

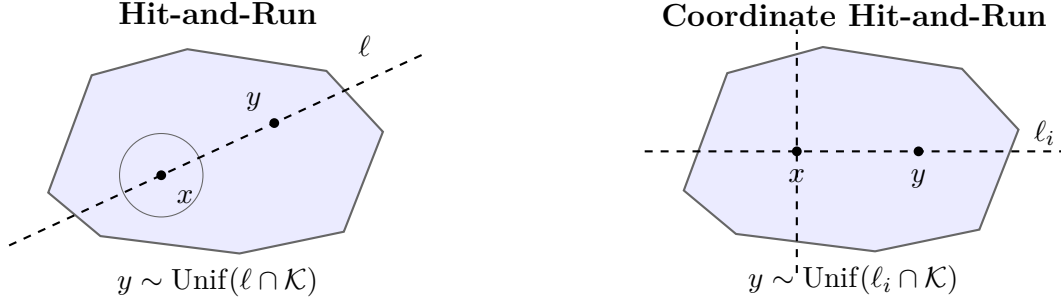


Figure 1.1: **Left:** A HAR kernel at $x \in K$ chooses a random line ℓ through x and samples y uniformly from the chord $\ell \cap K$. **Right:** A CHAR kernel chooses an axis-parallel line ℓ_i through x and samples y uniformly from the chord $\ell_i \cap K$.

cross-ratio distance, and the latter was already the best possible. Thus, it remained a tantalizing open problem to bridge the analysis of Hit-and-Run with improved isoperimetric constants.

Chen and Eldan [CE26] made a conceptually important breakthrough, showing that for isotropic convex bodies, the convergence rate is indeed nearly quadratic in the dimension, albeit with a polynomial dependence on initial warmness and the distance to the target. Interestingly, their proof technique, called a localization scheme [CE25], used a stochastic localization process to reduce the analysis from general convex bodies to that of a highly concentrated Gaussian restricted to a convex body; this general stochastic technique has also been the driving force for much of the progress in bounding the KLS constant. So, although they lost the logarithmic dependence on the warmness parameter, they showed that the conjectured dimension dependence is indeed plausible and valid in the isotropic setting.

Recent work [KVZ26, KV25b, KV25c] took a different approach inspired by continuous diffusion, and showed that the In-and-Out random walk has a convergence rate that matches the best-known bounds for the Ball walk, with stronger output guarantees and, importantly, a novel proof framework that directly connects convergence (in fact, per-step contraction towards the target) with functional isoperimetric constants of the target distribution. Nevertheless, In-and-Out also has an unavoidable linear dependence on the warmness parameter, highlighting the open problem about Hit-and-Run:

Can the convergence rate of Hit-and-Run be bounded by the Poincaré/KLS constants while maintaining its logarithmic dependence on the warmness parameter?

In this paper, we answer this question affirmatively for both Hit-and-Run and CHAR. The proof technique bypasses the traditional method of bounding the conductance and directly connects the convergence rate to functional isoperimetric constants.

The main high-level idea of the proof is to produce a dual certificate bounding the spectral gap. Rewriting the spectral gap first in terms of dual certificates leads to the Babuška–Aziz constant [BA72, HP83] studied in the analysis of PDEs; for bounded domains, this is known to be asymptotically bounded by the Improved Poincaré constant of the target density. The latter is a refined version of the classical Poincaré constant, incorporating distance to the boundary as a weight function for points in the domain [HS94, Cos17, Zsu20]. We show that this constant can be bounded in terms of the usual Poincaré constant for convex bodies resulting in a mixing time bound of $O(n^2 C_{\text{PI}} \log(M/\varepsilon))$. Applying this to CHAR, with the only change being in the first step of connecting the spectral gap to the Babuška–Aziz constant, results in a mixing time bound of $O(n^3 C_{\text{PI}} \log(M/\varepsilon))$, which is conjectured to be asymptotically tight. We note that under this new analytical framework, the proofs for Hit-and-Run and CHAR are essentially the same.

We discuss the proof techniques in more detail after stating the main results.

1.1 Results

Below, $C_{\text{PI}}(\pi)$ denotes the Poincaré constant of a probability measure π , formally defined in §1.2.

Theorem 1.1 (Spectral gaps). *Let $n \geq 2$ and $\mathcal{K} \subset \mathbb{R}^n$ be a bounded, open, and convex set with unit ball inside, and let π be the uniform distribution over $\mathcal{K}$. Then, the spectral gaps of Hit-and-Run and Coordinate Hit-and-Run satisfy*

$$\lambda_{\text{HR}} \gtrsim \frac{1}{n^2 C_{\text{PI}}(\pi)} \gtrsim \frac{1}{n^2 \|\text{cov } \pi\|_{\text{op}} \log n}, \quad \lambda_{\text{CHAR}} \gtrsim \frac{1}{n^3 C_{\text{PI}}(\pi)} \gtrsim \frac{1}{n^3 \|\text{cov } \pi\|_{\text{op}} \log n}.$$

Corollary 1.2 (Mixing time). *Under the assumptions of Theorem 1.1, let π_N^{HR} and π_N^{CHAR} be the distribution of the N -th iterate of Hit-and-Run and Coordinate Hit-and-Run initialized at π_0 , respectively. Then, for some universal constant $c > 0$*

$$\begin{aligned} \chi^2(\pi_N^{\text{HR}} \parallel \pi) &\leq \exp\left(-\frac{cN}{n^2 \|\text{cov } \pi\|_{\text{op}} \log n}\right) \chi^2(\pi_0 \parallel \pi), \\ \chi^2(\pi_N^{\text{CHAR}} \parallel \pi) &\leq \exp\left(-\frac{cN}{n^3 \|\text{cov } \pi\|_{\text{op}} \log n}\right) \chi^2(\pi_0 \parallel \pi). \end{aligned}$$

In particular, for given $\varepsilon > 0$, the number of iterations for Hit-and-Run and Coordinate Hit-and-Run to achieve ε -distance to π in χ^2 -divergence is bounded as

$$N_{\text{HR}} \lesssim n^2 \|\text{cov } \pi\|_{\text{op}} \log n \log \frac{\chi^2(\pi_0 \parallel \pi)}{\varepsilon}, \quad N_{\text{CHAR}} \lesssim n^3 \|\text{cov } \pi\|_{\text{op}} \log n \log \frac{\chi^2(\pi_0 \parallel \pi)}{\varepsilon}.$$

1.2 Technical overview

1.2.1 Preliminaries

We recall the functional-analytic facts used later, with more details in §A.

Basics. Throughout the paper, we use the same symbol to denote a probability measure and its Lebesgue density when it is clear from context, and $d\pi(x) = \frac{dx}{\text{vol } \mathcal{K}}$ denotes the uniform measure on $\mathcal{K} \subset \mathbb{R}^n$. We write $\langle f, g \rangle_\pi := \int_{\mathcal{K}} fg d\pi$, $\|f\|_2^2 := \langle f, f \rangle_\pi = \mathbb{E}_\pi[f^2]$, and $L_0^2(\pi) := \{f \in L^2(\pi) : \int_{\mathcal{K}} f d\pi = 0\}$, the set of centered L^2 functions. The mean and variance of $f \in L^2(\pi)$ are denoted as $\pi f := \mathbb{E}_\pi f = \int_{\mathcal{K}} f d\pi$ and $\text{Var}_\pi f = \|f - \pi f\|_2^2$.

A function $f \in L^2(\mathcal{K})$ has *weak derivative* $\partial_j f \in L^2(\mathcal{K})$ if $\int_{\mathcal{K}} f \partial_j \varphi dx = -\int_{\mathcal{K}} (\partial_j f) \varphi dx$ for every $\varphi \in C_c^\infty(\mathcal{K})$. The *Sobolev space* $H^1(\mathcal{K})$ consists of the L^2 functions whose weak derivatives belong to L^2 . Recall that it is a Hilbert space with $\langle f, g \rangle_{H^1} = \langle f, g \rangle_{L^2} + \langle \nabla f, \nabla g \rangle_{L^2}$. We denote the closure of compactly-supported smooth functions with respect to the H^1 -norm by $H_0^1(\mathcal{K}) := \overline{C_c^\infty(\mathcal{K})}^{H^1(\mathcal{K})}$. For a vector field $u = (u_1, \dots, u_n)$, we use $H^1(\mathcal{K}; \mathbb{R}^n) := H^1(\mathcal{K})^n$ and $\|\nabla u\|_2^2 = \sum_{i,j=1}^n \|\partial_j u_i\|_2^2$.

For $u \in H^1(\mathcal{K}; \mathbb{R}^n)$, its *weak divergence* is $\text{div } u := \sum_{i=1}^n \partial_i u_i \in L^2(\mathcal{K})$. Equivalently, $\int_{\mathcal{K}} (\text{div } u) \varphi dx = -\int_{\mathcal{K}} u \cdot \nabla \varphi dx$ for any $\varphi \in C_c^\infty(\mathcal{K})$. If $u \in H_0^1(\mathcal{K}; \mathbb{R}^n)$, approximation by compactly supported smooth vector fields gives $\int_{\mathcal{K}} \text{div } u d\pi = 0$.

Functional inequalities. For $\delta(x) := \text{dist}(x, \partial\mathcal{K}) = \min_{y \in \partial\mathcal{K}} |x - y|$ and a probability measure π over $\mathcal{K}$, its Poincaré constant and improved Poincaré constant are defined as the smallest constants satisfying that for any test function $f \in H^1(\mathcal{K})$,

$$\text{Var}_\pi f \leq C_{\text{PI}}(\pi) \int_{\mathcal{K}} |\nabla f|^2 d\pi, \tag{PI}$$

$$\text{Var}_\pi f \leq C_{\text{IPI}}(\pi) \int_{\mathcal{K}} \delta^2 |\nabla f|^2 d\pi. \tag{IPI}$$

Dirichlet form and spectral gap. Let $\mathcal{G}$ be a sub- σ -algebra of the Borel σ -algebra on $\mathcal{K}$. Recall that the conditional expectation $P_{\mathcal{G}}f := \mathbb{E}_{\pi}[f \mid \mathcal{G}]$ is the orthogonal projection of $L^2(\pi)$ onto the closed subspace of $\mathcal{G}$ -measurable functions:

$$P_{\mathcal{G}}^* = P_{\mathcal{G}}, \quad P_{\mathcal{G}}^2 = P_{\mathcal{G}}, \quad \langle f, (\text{Id} - P_{\mathcal{G}})f \rangle_{\pi} = \|f - P_{\mathcal{G}}f\|_2^2. \quad (1.1)$$

For a π -reversible Markov operator P , its Dirichlet form is $\mathcal{E}_P(f, f) := \langle f, (\text{Id} - P)f \rangle_{\pi}$, and the spectral gap of P is defined as

$$\lambda_P := \inf_{0 \neq f \in L_0^2(\pi)} \frac{\mathcal{E}_P(f, f)}{\|f\|_2^2}.$$

A positive spectral gap of positive semidefinite P implies contraction in χ^2 -divergence: if $q_0 = \frac{d\mu_0}{d\pi}$ belongs to $L^2(\pi)$ and $\mu_N = \mu_0 P^N$,

$$\chi^2(\mu_N \parallel \pi) = \|P^N(q_0 - 1)\|_2^2 \leq (1 - \lambda_P)^{2N} \chi^2(\mu_0 \parallel \pi) \leq \exp(-2\lambda_P N) \chi^2(\mu_0 \parallel \pi). \quad (1.2)$$

1.2.2 Proof ideas

We will define a bounded linear operator $T : F \rightarrow G$ for two Hilbert spaces F and G so that $\|f\|_F^2$ and $\|Tf\|_G^2$ correspond to $\|f\|_2^2$ and $\mathcal{E}_P(f, f)$ in our setting, respectively. Then, our goal is to establish $\|Tf\|_G^2 \geq \lambda \|f\|_F^2$ for some $\lambda > 0$. A natural *dual* approach is to find, for each $f \in F$, a dual-certificate $g_f \in G$ such that $T^*g_f = f$ for the adjoint $T^* : G \rightarrow F$. If the certificate also satisfies $\|g_f\|_G^2 \leq C \|f\|_F^2$, then

$$\|f\|_F^2 = \langle f, f \rangle_F = \langle f, T^*g_f \rangle_F = \langle Tf, g_f \rangle_G \leq \|Tf\|_G \|g_f\|_G \leq \sqrt{C} \|Tf\|_G \|f\|_F,$$

which implies $\|f\|_F^2 \leq C \|Tf\|_G^2$, and it leads to $\lambda \geq C^{-1}$. In summary, we should tackle two concrete problems: (i) find the dual certificate g_f such that $T^*g_f = f$, and (ii) show $\|g_f\|_G^2 \leq C \|f\|_F^2$ for some $C > 0$.

We now illustrate the main idea through Coordinate Hit-and-Run, whose finite product structure makes the adjoint transparent. Fix an orthonormal basis $e_1, \dots, e_n$, and let P_i be conditional expectation given all coordinates except the i -th one; equivalently, $P_i f$ is the uniform average of f on each e_i -parallel chord. By (1.1),

$$P_{\text{CHAR}} = \frac{1}{n} \sum_{i=1}^n P_i, \\ \mathcal{E}_{\text{CHAR}}(f, f) := \langle f, (\text{Id} - P_{\text{CHAR}})f \rangle_{\pi} = \frac{1}{n} \sum_{i=1}^n \|f - P_i f\|_2^2.$$

In this case, we can take $F := L_0^2(\pi)$ and $G := \{g = (g_1, \dots, g_n) : g_i \in L^2(\pi)\}$ with norm $\|f\|_F^2 = \|f\|_2^2$ and $\|g\|_G^2 = \frac{1}{n} \sum_{i=1}^n \|g_i\|_2^2$. Also, we define $T : F \rightarrow G$ by $(Tf)_i = (\text{Id} - P_i)f$ for $i \in [n]$, so $\|Tf\|_G^2 = \mathcal{E}_{\text{CHAR}}(f, f)$ as desired. Let us now compute the adjoint T^* . For $g = (g_1, \dots, g_n) \in G$, using $P_i = P_i^*$,

$$\langle Tf, g \rangle_G = \frac{1}{n} \sum_i \langle (\text{Id} - P_i)f, g_i \rangle_{\pi} = \frac{1}{n} \sum_i \langle f, (\text{Id} - P_i)g_i \rangle_{\pi}.$$

Hence, $T^*g = \frac{1}{n} \sum_i (\text{Id} - P_i)g_i$.

With the dual-certificate approach in mind, we address the first problem of finding g such that $T^*g = f$. To this end, we could consider two sufficient conditions given as

$$P_i g_i = 0 \quad \text{for } i \in [n] \quad \& \quad \frac{1}{n} \sum_{i=1}^n g_i = f.$$

A convenient way to enforce the first condition is to set $g_i = n \partial_i u_i$ for each $i \in [n]$ for some $u \in H_0^1(\mathcal{K}; \mathbb{R}^n)$ extended by zero outside of $\mathcal{K}$. Indeed, for a.e. $z \in e_i^\perp$, if $I_z := \{s \in \mathbb{R} : z + se_i \in \mathcal{K}\}$, then for $t \in I_z$, the fundamental theorem of calculus leads to

$$(P_i g_i)(z + te_i) = \frac{n}{|I_z|} \int_{I_z} \partial_i u_i(z + se_i) ds = 0.$$

Then, the second condition corresponds to $\operatorname{div} u = f$. In summary, the first problem comes down to finding $u \in H_0^1(\mathcal{K}; \mathbb{R}^n)$ such that $\operatorname{div} u = f$.

We now address the second problem of showing $\|g\|_G^2 \leq C \|f\|_F^2$ for some C . Since

$$\|g\|_G^2 = \frac{1}{n} \sum_{i=1}^n \|g_i\|_2^2 = n \sum_{i=1}^n \|\partial_i u_i\|_2^2 \leq n \|\nabla u\|_2^2,$$

the second problem can be reduced to the following optimization problem: among functions u with $\operatorname{div} u = f$, find a function u that minimizes $\|\nabla u\|_2^2$. Namely, one should minimize the Frobenius norm $\|\nabla u\|_2$ with trace constraint $\operatorname{div} u = \operatorname{tr}(\nabla u) = f$. This is where the Babuška–Aziz constant comes in:

$$C_{\text{BA}}(\pi) := \sup_{0 \neq f \in L_0^2(\pi)} \inf_{\substack{u \in H_0^1(\mathcal{K}; \mathbb{R}^n) \\ \operatorname{div} u = f}} \frac{\|\nabla u\|_2^2}{\|f\|_2^2}.$$

The minimizer satisfies $\operatorname{div} u = f$ and $\|\nabla u\|_2^2 \leq C_{\text{BA}}(\pi) \|f\|_2^2$. Consequently, $\|g\|_G^2 \leq n C_{\text{BA}}(\pi) \|f\|_F^2$, and the dual argument gives $\lambda_{\text{CHAR}} \geq \frac{1}{n C_{\text{BA}}(\pi)}$. We formalize this functional analytic result as follows.

Lemma 1.3. *Suppose $C_{\text{BA}}(\pi) < \infty$. Then, for every $f \in L_0^2(\pi)$, there exists $u \in H_0^1(\mathcal{K}; \mathbb{R}^n)$ such that*

$$\operatorname{div} u = f, \quad \text{and} \quad \|\nabla u\|_2^2 \leq C_{\text{BA}}(\pi) \|f\|_2^2. \quad (1.3)$$

Proof. Finiteness of $C_{\text{BA}}(\pi)$ makes the feasible set in the inner infimum nonempty. Equip $H_0^1(\mathcal{K}; \mathbb{R}^n)$ with the norm $\|\nabla u\|_2$. By the Poincaré inequality for $H_0^1(\mathcal{K}; \mathbb{R}^n)$, this is an equivalent Hilbert norm on $H_0^1(\mathcal{K}; \mathbb{R}^n)$. Since $\|\operatorname{div} u\|_2 \leq \sqrt{n} \|\nabla u\|_2$, it follows that $\operatorname{div} : H_0^1(\mathcal{K}; \mathbb{R}^n) \rightarrow L_0^2(\pi)$ is a bounded linear operator (so continuous). Hence, $\operatorname{div}^{-1}(\{f\}) = \{u \in H_0^1(\mathcal{K}; \mathbb{R}^n) : \operatorname{div} u = f\}$ is a nonempty closed convex subset of $H_0^1(\mathcal{K}; \mathbb{R}^n)$. By the Hilbert projection theorem, the infimum in the definition of $C_{\text{BA}}(\pi)$ is attained, and its minimizer satisfies (1.3). $\square$

In §2, we formally bound the spectral gaps of HAR and CHAR in terms of the Babuška–Aziz constant of the target distribution. Notably, the spectral-gap bound for CHAR is smaller by a factor of the dimension. The rest of the analysis is independent of the Markov chain and will apply to both HAR and CHAR. The next step is the following result from [Zsu20] which relates the Babuška–Aziz constant to the Improved Poincaré constant. We give a self-contained proof of this in §B.

Theorem 1.4. *For a bounded convex domain in $\mathbb{R}^n$, we have $C_{\text{BA}} \leq 1 + 4C_{\text{IP}}$.*

Then, in §3, we show that $C_{\text{IP}}(\pi) \lesssim n^2 C_{\text{PI}}(\pi)$ for a convex body containing a unit ball (Theorem 3.1 and Corollary 3.2). These ingredients suffice to prove the main results.

1.3 Proof of main result

Proof of Theorem 1.1. Note that $C_{\text{PI}}(\pi) \geq \|\text{cov } \pi\|_{\text{op}} \gtrsim 1/n$ when $B(0, 1) \subset \mathcal{K}$ (see [KV26, Lemma 2.1]). By applying Propositions 2.1 and 2.2 with Theorem 1.4 ($C_{\text{BA}} \leq 1 + 4C_{\text{PI}}$) and Corollary 3.2 ($C_{\text{PI}} \lesssim n^2 C_{\text{PI}}/r^2$ with $r = 1$),

$$\lambda_{\text{HR}} \gtrsim \frac{1}{n^2 C_{\text{PI}}(\pi)}, \quad \lambda_{\text{CHAR}} \gtrsim \frac{1}{n^3 C_{\text{PI}}(\pi)}.$$

Using $C_{\text{PI}}(\pi) \lesssim \|\text{cov } \pi\|_{\text{op}} \log n$ [Kla23], we complete the proof. $\square$

Proof of Corollary 1.2. This follows from the spectral-gap results in Theorem 1.1 and (1.2). $\square$

2 Spectral gap via Babuška–Aziz constant

2.1 Hit-and-Run

Let σ be uniform probability measure on $\mathbb{S}^{n-1}$. For $\theta \in \mathbb{S}^{n-1}$, let P_θ be conditional expectation with respect to the orthogonal projection onto $\theta^\perp$, so $P_\theta f$ is the uniform average of f on each θ -parallel chord. Then,

$$P_{\text{HR}} = \int_{\mathbb{S}^{n-1}} P_\theta \, d\sigma(\theta),$$

$$\mathcal{E}_{\text{HR}}(f, f) = \langle f, (\text{Id} - P_{\text{HR}})f \rangle_\pi = \int_{\mathbb{S}^{n-1}} \|f - P_\theta f\|_2^2 \, d\sigma(\theta).$$

For θ drawn from $\mathbb{S}^{n-1}$, the natural dual certificate is $g_\theta(x) := n \theta^\top \nabla u(x) \theta$.

Proposition 2.1. *If $C_{\text{BA}}(\pi) < \infty$, then*

$$\lambda_{\text{HR}} \geq \frac{n+2}{n(2+C_{\text{BA}}(\pi))} \gtrsim \frac{1}{1+C_{\text{BA}}(\pi)}.$$

Proof. Fix $f \in L_0^2(\pi)$, and pick $u \in H_0^1(\mathcal{K}; \mathbb{R}^n)$ given by Lemma 1.3 that satisfies $\text{div } u = f$ and $\|\nabla u\|_2^2 \leq C_{\text{BA}}(\pi) \|f\|_2^2$. For $\theta \in \mathbb{S}^{n-1}$, define

$$g_\theta(x) := n \theta^\top \nabla u(x) \theta \quad \text{for a.e. } x \in \mathcal{K}.$$

Let $u \in H_0^1(\mathcal{K}; \mathbb{R}^n)$ and choose $u_k \in C_c^\infty(\mathcal{K}; \mathbb{R}^n)$ converging to u in H^1 . Define $g_{\theta,k} := n \theta^\top \nabla u_k \theta$. The fundamental theorem of calculus on every θ -parallel line gives $P_\theta g_{\theta,k} = 0$, because u_k is compactly supported. Since $g_{\theta,k} \rightarrow g_\theta$ in $L^2(\pi)$ and conditional expectation is an L^2 contraction, $P_\theta g_\theta = \lim_{k \rightarrow \infty} P_\theta g_{\theta,k} = 0$ in L^2 . Hence, $0 = \langle f, P_\theta g_\theta \rangle_\pi = \langle P_\theta f, g_\theta \rangle_\pi$.

Recall that for a square matrix M , rotational invariance of $d\sigma(\theta)$ gives

$$\mathbb{E}_\theta[\theta \theta^\top] = \frac{1}{n} \text{Id}, \tag{2.1}$$

$$\mathbb{E}_\theta[(\theta^\top M \theta)^2] = \frac{(\text{tr } M)^2 + 2 \|\text{sym } M\|_{\text{F}}^2}{n(n+2)}, \tag{2.2}$$

where the second identity follows from $\mathbb{E}[\theta_i \theta_j \theta_k \theta_\ell] = \frac{\delta_{ij} \delta_{k\ell} + \delta_{ik} \delta_{j\ell} + \delta_{i\ell} \delta_{jk}}{n(n+2)}$. By (2.1) and $\text{div } u = f$,

$$\int_{\mathbb{S}^{n-1}} g_\theta \, d\sigma(\theta) = \text{tr } \nabla u = f.$$

Therefore, using $\langle P_\theta f, g_\theta \rangle_\pi = 0$ and Cauchy-Schwarz,

$$\begin{aligned} \|f\|_2^2 &= \int_{\mathbb{S}^{n-1}} \langle f, g_\theta \rangle_\pi d\sigma(\theta) = \int_{\mathbb{S}^{n-1}} \langle f - P_\theta f, g_\theta \rangle_\pi d\sigma(\theta), \\ \|f\|_2^4 &\leq \mathcal{E}_{\text{HR}}(f, f) \int_{\mathbb{S}^{n-1}} \|g_\theta\|_2^2 d\sigma(\theta). \end{aligned} \quad (2.3)$$

For a square matrix M , we denote its symmetrization by $\text{sym } M := (M + M^\top)/2$. By integration by parts, for $u \in H_0^1(\mathcal{K}; \mathbb{R}^n)$,

$$2 \|\text{sym } \nabla u\|_2^2 = \|\nabla u\|_2^2 + \|\text{div } u\|_2^2, \quad (2.4)$$

since the cross term satisfies $\sum_{i,j} \int_{\mathcal{K}} (\partial_j u_i)(\partial_i u_j) d\pi = \int_{\mathcal{K}} (\text{div } u)^2 d\pi$. Applying (2.2) to $M = \nabla u$, and then using (2.4), $\text{div } u = f$, and $\|\nabla u\|_2^2 \leq C_{\text{BA}}(\pi) \|f\|_2^2$, we obtain

$$\begin{aligned} \int_{\mathbb{S}^{n-1}} \|g_\theta\|_2^2 d\sigma(\theta) &= \frac{n}{n+2} \mathbb{E}_\pi[(\text{div } u)^2 + 2 \|\text{sym } \nabla u\|_2^2] = \frac{n}{n+2} (\|f\|_2^2 + 2 \|\text{sym } \nabla u\|_2^2) \\ &\leq \frac{n(2 + C_{\text{BA}}(\pi))}{n+2} \|f\|_2^2. \end{aligned}$$

Substitution into (2.3) gives

$$\mathcal{E}_{\text{HR}}(f, f) \geq \frac{n+2}{n(2 + C_{\text{BA}}(\pi))} \|f\|_2^2.$$

Taking the infimum over nonzero $f \in L_0^2(\pi)$ completes the proof. $\square$

2.2 Coordinate Hit-and-Run

Proposition 2.2. *If $C_{\text{BA}}(\pi) < \infty$, then*

$$\lambda_{\text{CHAR}} \geq \frac{1}{nC_{\text{BA}}(\pi)}.$$

Proof. Fix $f \in L_0^2(\pi)$ and let u be given by Lemma 1.3. For $i = 1, \dots, n$, set $g_i := n \partial_i u_i$ and note that $\frac{1}{n} \sum_{i=1}^n g_i = \text{div } u = f$. A similar argument in the proof of Proposition 2.1 gives $P_i g_i = 0$, and thus $0 = \langle f, P_i g_i \rangle_\pi = \langle P_i f, g_i \rangle_\pi$. Then,

$$\begin{aligned} \|f\|_2^2 &= \frac{1}{n} \sum_{i=1}^n \langle f, g_i \rangle_\pi = \frac{1}{n} \sum_{i=1}^n \langle f - P_i f, g_i \rangle_\pi, \\ \|f\|_2^4 &\leq \mathcal{E}_{\text{CHAR}}(f, f) \frac{1}{n} \sum_{i=1}^n \|g_i\|_2^2 = n \mathcal{E}_{\text{CHAR}}(f, f) \sum_{i=1}^n \|\partial_i u_i\|_2^2. \end{aligned} \quad (2.5)$$

By Lemma 1.3,

$$\sum_{i=1}^n \|\partial_i u_i\|_2^2 \leq \|\nabla u\|_2^2 \leq C_{\text{BA}}(\pi) \|f\|_2^2.$$

Substitution into (2.5) gives $\mathcal{E}_{\text{CHAR}}(f, f) \geq \frac{1}{nC_{\text{BA}}(\pi)} \|f\|_2^2$, and taking the infimum over nonzero $f \in L_0^2(\pi)$ finishes the proof. $\square$

3 Bounding the improved Poincaré constant

Since $C_{\text{BA}} \leq 1 + 4C_{\text{PI}}$ for a bounded convex domain (Theorem 1.4), we are only left with bounding the improved Poincaré constant (IPI) in terms of the usual Poincaré constant. Before we proceed, we briefly survey prior work on (IPI). Improved Poincaré inequalities with the distance to the boundary as a weight have been studied for John domains (which include convex domains), starting with [HS94] and further developed in [DD08, CW10]. In particular, for convex domains, [CW10] gives a quantitative bound in terms of the *eccentricity* of the domain. In our notation, this implies $C_{\text{PI}}(\pi) \leq C(n) R^2/r^2$, where R and r denote the outer-radius and in-radius of the domain, respectively, and the dimension dependence is absorbed into $C(n)$. For our application, we need a bound in terms of the ordinary Poincaré constant with explicit dimension dependence, so in Corollary 3.2 we show $C_{\text{IPI}}(\pi) \lesssim n^2 C_{\text{PI}}(\pi)/r^2$.

Theorem 3.1. *For any locally Lipschitz function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and a convex body $\mathcal{K} \subset \mathbb{R}^n$ containing a ball of radius r , we have*

$$\inf_z \int_{\mathcal{K}} |f(x) - z| dx \lesssim \frac{n\sqrt{C_{\text{PI}}(\pi)}}{r} \int_{\mathcal{K}} \delta(x) |\nabla f| dx.$$

This gives the following immediate corollary:

Corollary 3.2. *For a convex body $\mathcal{K}$ in $\mathbb{R}^n$ whose maximum inscribed ball has radius r , we have*

$$C_{\text{IPI}}(\pi) \lesssim \frac{n^2 C_{\text{PI}}(\pi)}{r^2}.$$

Proof. Since $C^\infty(\mathbb{R}^n)|_{\mathcal{K}}$ is dense in $H^1(\mathcal{K})$, it suffices to prove the claim for $f \in C^\infty(\mathbb{R}^n)|_{\mathcal{K}}$. Then, Theorem 3.1 holds with z set to the median of f , i.e., for any $g : \mathbb{R}^n \rightarrow \mathbb{R}$ with median 0, we have

$$\int_{\mathcal{K}} |g(x)| dx \lesssim \frac{n\sqrt{C_{\text{PI}}(\pi)}}{r} \int_{\mathcal{K}} \delta(x) |\nabla g| dx.$$

Now set $u(x) := (f(x) - z_f)^+$ where z_f is the median of f and apply the above to $g = u^2$. Then, 0 is the median of g . Denoting $C := nC_{\text{PI}}(\pi)^{1/2}/r$, we have

$$\int_{\mathcal{K}} u^2 dx \lesssim C \int_{\mathcal{K}} \delta(x) u |\nabla u| dx \leq C \left(\int_{\mathcal{K}} u^2 dx \right)^{1/2} \left(\int_{\mathcal{K}} \delta(x)^2 |\nabla u|^2 dx \right)^{1/2}.$$

Hence, $\int_{\mathcal{K}} u^2 dx \lesssim C^2 \int_{\mathcal{K}} \delta(x)^2 |\nabla u|^2 dx$.

We get the same conclusion with $v = (z_f - f(x))^+$ in place of u . Adding these,

$$\text{Var}_\pi f \leq \int_{\mathcal{K}} (f - z_f)^2 d\pi \lesssim \frac{n^2 C_{\text{PI}}(\pi)}{r^2} \int_{\mathcal{K}} \delta(x)^2 |\nabla f|^2 d\pi,$$

which completes the proof. $\square$

We now give an elementary proof of the theorem.

Proof of Theorem 3.1. We may assume that $B(0, r) \subset \mathcal{K}$ by translation. The idea is just that since $\mathcal{K}$ contains a ball of radius r , we will consider a slight contraction of $\mathcal{K}$ by roughly $(1 - 1/n)$. Set $\alpha = 1 - \frac{1}{2n}$, and note that

$$\inf_z \int_{\mathcal{K}} |f(x) - z| dx \leq \int_{\mathcal{K}} |f(x) - f(\alpha x)| dx + \inf_z \int_{\mathcal{K}} |f(\alpha x) - z| dx.$$

For the first term, since the diameter of isotropic convex bodies is at most $\sqrt{n(n+2)} \leq n+1$ [KLS95, Theorem 4.1], we have $\sup_{\mathcal{K}} |x| \lesssim n \|\text{cov } \pi\|_{\text{op}}^{1/2} \leq n C_{\text{PI}}(\pi)^{1/2}$. Hence, for the gauge function $\|x\|_{\mathcal{K}} := \inf\{t > 0 : x \in t\mathcal{K}\}$,

$$\begin{aligned} \int_{\mathcal{K}} |f(x) - f(\alpha x)| dx &\leq \int_{\mathcal{K}} \int_{\alpha}^1 |x| |\nabla f(tx)| dt dx \lesssim n \sqrt{C_{\text{PI}}} \int_{\alpha}^1 t^{-n} \int_{t\mathcal{K}} |\nabla f(y)| dy dt \\ &= n \sqrt{C_{\text{PI}}} \int_{\mathcal{K}} |\nabla f(y)| \int_{\max\{\alpha, \|y\|_{\mathcal{K}}\}}^1 t^{-n} dt dy \\ &\leq n \sqrt{C_{\text{PI}}} \int_{\mathcal{K}} |\nabla f(y)| \alpha^{-n} \int_{\max\{\alpha, \|y\|_{\mathcal{K}}\}}^1 dt dy \\ &\leq n \sqrt{C_{\text{PI}}} \int_{\mathcal{K}} |\nabla f(y)| \cdot 2(1 - \|y\|_{\mathcal{K}}) dy \\ &\lesssim \frac{n \sqrt{C_{\text{PI}}}}{r} \int_{\mathcal{K}} \delta(y) |\nabla f(y)| dy, \end{aligned}$$

where in the last line, we used $\delta(y) \geq r(1 - \|y\|_{\mathcal{K}})$, since $\|y\|_{\mathcal{K}} \mathcal{K} + (1 - \|y\|_{\mathcal{K}}) B(0, r) \subset \mathcal{K}$.

For the second term, using the standard equivalence between the L^1 and L^2 -Poincaré constants for logconcave measures, and noting that the constant is $\alpha^2 C_{\text{PI}}$ for $\alpha\mathcal{K}$,

$$\inf_z \int_{\mathcal{K}} |f(\alpha x) - z| dx = \alpha^{-n} \inf_z \int_{\alpha\mathcal{K}} |f(x) - z| dx \leq \alpha^{-n} \alpha \sqrt{C_{\text{PI}}} \int_{\alpha\mathcal{K}} |\nabla f| dx \lesssim \frac{n \sqrt{C_{\text{PI}}}}{r} \int_{\alpha\mathcal{K}} \delta(x) |\nabla f| dx,$$

where we used $\delta(x) \geq \frac{r}{2n}$ for $x \in \alpha\mathcal{K}$. The theorem follows by adding these two bounds. $\square$

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A Functional-analytic background

This appendix expands the functional-analytic facts used in the proof. All Sobolev spaces are taken over the open set $\mathcal{K}$. A standard reference is Evans [Eva10, Chapter 5].

Hilbert spaces. A real *Hilbert space* is a vector space F with an inner product $\langle f, g \rangle_F$ that is complete in the induced norm $\|f\|_F^2 := \langle f, f \rangle_F$, where completeness means that every Cauchy sequence converges to an element of the same space. If $T : F \rightarrow G$ is a bounded linear map between Hilbert spaces, its adjoint is the unique bounded map $T^* : G \rightarrow F$ satisfying

$$\langle Tf, g \rangle_G = \langle f, T^*g \rangle_F \quad \text{for every } f \in F, g \in G.$$

The Hilbert projection theorem states that every nonempty closed convex set contains a unique element of minimum norm.

Weak derivatives. For smooth f and $\varphi \in C_c^\infty(\mathcal{K})$, integration by parts gives

$$\int_{\mathcal{K}} f \partial_j \varphi \, dx = - \int_{\mathcal{K}} (\partial_j f) \varphi \, dx.$$

There is no boundary term because φ is compactly supported in $\mathcal{K}$. We take this identity as the definition for $f \in L^2(\mathcal{K})$ that need not possess a classical derivative: a function $v_j \in L^2(\mathcal{K})$ is the *weak derivative* $\partial_j f$ if

$$\int_{\mathcal{K}} f \partial_j \varphi \, dx = - \int_{\mathcal{K}} v_j \varphi \, dx \quad \text{for every } \varphi \in C_c^\infty(\mathcal{K}).$$

Thus, the definition preserves precisely the integration-by-parts rule needed later. The weak derivative is unique (in a.e. sense). It is also stable under L^2 limits: if $f_k \rightarrow f$ and $\partial_j f_k \rightarrow v_j$ in $L^2(\mathcal{K})$, passing to the limit in the preceding identity shows that $\partial_j f = v_j$.

Sobolev spaces. The Sobolev space is

$$H^1(\mathcal{K}) := \{f \in L^2(\mathcal{K}) : \partial_j f \in L^2(\mathcal{K}) \text{ for every } j \in [n]\} \quad \text{with norm } \|f\|_{H^1(\mathcal{K})}^2 := \int_{\mathcal{K}} (|f|^2 + |\nabla f|^2) \, dx.$$

The stability of weak derivatives under limits shows that $H^1(\mathcal{K})$ is complete, hence Hilbert.

Let us consider the H^1 -closure of smooth functions with compact support in $\mathcal{K}$, $H_0^1(\mathcal{K}) := \overline{C_c^\infty(\mathcal{K})}^{H^1(\mathcal{K})}$. Namely, for any $f \in H_0^1(\mathcal{K})$, there exists a sequence of compactly-supported smooth functions φ_k such that $\varphi_k \rightarrow f$ in the H^1 norm. This closure encodes a zero boundary condition without requiring pointwise boundary values.

Actions on functions and measures. Let $\mathcal{B}(\mathcal{K})$ be the Borel σ -algebra of $\mathcal{K}$. A *Markov kernel* is a map $P : \mathcal{K} \times \mathcal{B}(\mathcal{K}) \rightarrow [0, 1]$ such that $P(x, \cdot)$ is a probability measure for every $x \in \mathcal{K}$ and $x \mapsto P(x, A)$ is measurable for every $A \in \mathcal{B}(\mathcal{K})$. For a bounded measurable function f and a probability measure μ , define

$$(Pf)(x) := \int_{\mathcal{K}} f(y) P(x, dy), \quad (\mu P)(A) := \int_{\mathcal{K}} P(x, A) d\mu(x).$$

The first action is the associated *Markov operator* on functions, while the second one is the law after one transition from an initial law μ .

Stationarity and reversibility. A probability measure π is *stationary* if $\pi P = \pi$. In this case $\pi(Pf) = \pi f$, so P preserves $L^2_0(\pi)$. Jensen's inequality and stationarity give

$$\|Pf\|_2^2 \leq \int_{\mathcal{K}} P(f^2) d\pi = \int_{\mathcal{K}} f^2 d\pi = \|f\|_2^2,$$

so P is a bounded linear map. Then, its adjoint P^* is characterized by $\langle Pf, g \rangle_{\pi} = \langle f, P^*g \rangle_{\pi}$ for every $f, g \in L^2(\pi)$. The operator is *self-adjoint* if $P = P^*$. A Markov kernel P is *reversible* with respect to π if $d\pi(x) P(x, dy) = d\pi(y) P(y, dx)$. This is equivalent to self-adjointness of P on $L^2(\pi)$.

B From Babuška–Aziz constant to improved Poincaré constant

Throughout this section, we use $\Omega \subset \mathbb{R}^n$ for the domain, $Q = L^2_0(\Omega)$ for the space of square-integrable functions over Ω with zero mean and norm $\|f\|_Q := \|f\|_2$, and $V = H^1_0(\Omega; \mathbb{R}^n)$ for the space of vector fields on Ω with square-integrable derivatives that vanish on the boundary and norm $\|v\|_V = \|\nabla v\|_2$ (note that this is the L^2 -norm of the Frobenius norm of $\nabla v \in \mathbb{R}^{n \times n}$). As implied in §1.2.2, we will focus on the divergence operator $\operatorname{div} : V \rightarrow Q$. Since $\int_{\Omega} \operatorname{div} v = \int_{\partial\Omega} v \cdot \nu = 0$ for $v \in V$ (here, ν is the outward normal vector), this divergence operator is well-defined.

In this section, we will recap the proof of Theorem 1.4 for the reader's convenience, focusing on Euclidean domains. Note that this result follows from combining [Cos17, Theorem 2.1] and [Zsu20, Lemma 3.4]. The proof of the theorem will follow from the following lemmas. The first one is just a dual definition of C_{BA} .

Lemma B.1 (Duality). *The following are equivalent for any $C > 0$:*

1. *For any $f \in Q$, there exists $v \in V$ such that $\operatorname{div} v = f$ and $\|\nabla v\|_2^2 \leq C \|f\|_2^2$.*
2. *For any $f \in Q$, it holds that $\|f\|_2^2 \leq C (\sup_{v \in V, v \neq 0} \frac{\langle f, \operatorname{div} v \rangle_Q}{\|\nabla v\|_2})^2$.*

When solving $\operatorname{div} v = f$, the Babuška–Aziz constant essentially asks how large $\|v\|_V$ is compared with $\|f\|_Q$, and this is the first statement (i.e., $\|v\|_V \leq C^{1/2} \|\operatorname{div} v\|_Q$ for some $v \in V$). Consider the adjoint $\operatorname{div}^* : Q \rightarrow V$ such that $\langle f, \operatorname{div} v \rangle_Q = \langle \operatorname{div}^* f, v \rangle_V$. Then, the RHS in the second statement is equivalent to $\|f\|_Q \leq C^{1/2} \|\operatorname{div}^* f\|_V$.

The next lemma derives an explicit solution of the supremum on the RHS in the previous lemma using the Riesz representation theorem.

Lemma B.2 (Map to vector field). *For any $f \in Q$, there is a unique $u \in V$ such that $\langle f, \operatorname{div} v \rangle_Q = \langle u, v \rangle_V$ for any $v \in V$, and*

$$\|\nabla u\|_2 = \sup_{v \in V, v \neq 0} \frac{\langle f, \operatorname{div} v \rangle_Q}{\|\nabla v\|_2}.$$

Proof. The map $v \mapsto \langle f, \operatorname{div} v \rangle$ is a bounded linear map, since $|\langle f, \operatorname{div} v \rangle| \leq \|f\|_2 \|\operatorname{div} v\|_2 \leq \sqrt{n} \|f\|_2 \|\nabla v\|_2$. By the Riesz representation theorem, there is a unique $u \in V$ such that

$$\langle f, \operatorname{div} v \rangle_Q = \langle u, v \rangle_V = \langle \nabla u, \nabla v \rangle \leq \|\nabla u\|_2 \|\nabla v\|_2.$$

Hence, the supremum on the RHS in the lemma statement is at most $\|\nabla u\|_2$. Setting $v = u$ shows equality. $\square$

Due to this lemma, it now suffices to show $\|f\|_Q \leq C^{1/2} \|u\|_V$ for some C . The next lemma is a calculus identity about divergence of vector fields.

Lemma B.3. *For $u \in V$, define the $n \times n$ matrix function G as $G_{ij} := \partial_j u_i - \partial_i u_j$ with $\|G\|^2 := \int_{\Omega} \sum_{i < j} G_{ij}^2 dx$. Then, $\|\nabla u\|_2^2 = \|\operatorname{div} u\|_2^2 + \|G\|^2$.*

Proof. Using integration by parts twice and the fact that partial derivatives commute,

$$\|\operatorname{div} u\|_2^2 = \sum_{ij} \int_{\Omega} \partial_i u_i \partial_j u_j = - \sum_{ij} \int_{\Omega} u_i \partial_i \partial_j u_j = \sum_{ij} - \int_{\Omega} u_i \partial_j \partial_i u_j = \sum_{ij} \int_{\Omega} \partial_j u_i \partial_i u_j.$$

Let us now compute $\|G\|^2$:

$$\begin{aligned} \|G\|^2 &= \sum_{i < j} \int_{\Omega} (\partial_j u_i - \partial_i u_j)^2 = \frac{1}{2} \sum_{ij} \int_{\Omega} (\partial_j u_i - \partial_i u_j)^2 = \sum_{ij} \int_{\Omega} (\partial_i u_j)^2 - \sum_{ij} \int_{\Omega} \partial_j u_i \partial_i u_j \\ &= \|\nabla u\|_2^2 - \|\operatorname{div} u\|_2^2, \end{aligned}$$

which completes the proof. $\square$

Next, using this definition of G , we get useful identities.

Lemma B.4. *For $f \in Q$, let $u \in V$ be the vector field given by Lemma B.2 and G be the matrix function for u defined in Lemma B.3. Then for $h := f - \operatorname{div} u$, we have*

1. $\int_{\Omega} h = 0$;
2. $\nabla h = \operatorname{div} G$ (i.e., $\partial_i h = \sum_{j=1}^n \partial_j G_{ij}$ for $i \in [n]$);
3. $\langle h, \operatorname{div} u \rangle_Q = \|G\|^2$.

Proof. Since $u = 0$ on $\partial\Omega$, we have $\int_{\Omega} \operatorname{div} u = \int_{\partial\Omega} \langle u, \nu \rangle = 0$. Hence, $\int_{\Omega} h = \int_{\Omega} (f - \operatorname{div} u) = 0$.

Next, using weak derivatives and integration by parts, we can deduce $\Delta u = \nabla f$ from $\langle f, \operatorname{div} v \rangle_Q = \langle u, v \rangle_V$. Using this, for each i ,

$$\sum_j \partial_j G_{ij} = \sum_j \partial_j (\partial_j u_i - \partial_i u_j) = \Delta u_i - \partial_i \operatorname{div} u = \partial_i f - \partial_i \operatorname{div} u = \partial_i h.$$

Taking $v = u$ in the definition of u , we obtain $\langle f, \operatorname{div} u \rangle_Q = \|\nabla u\|_2^2 = \|\operatorname{div} u\|_2^2 + \|G\|^2$, and this implies that $\langle h, \operatorname{div} u \rangle_Q = \langle f - \operatorname{div} u, \operatorname{div} u \rangle_Q = \|G\|^2$. $\square$

The following lemma captures the heart of the argument. Recall that $\delta(x)$ is the distance of x to the boundary of Ω .

Lemma B.5. *For any $h \in Q$ and G an anti-symmetric matrix function satisfying $\nabla h = \operatorname{div} G$, we have $\|\delta \nabla h\|_2 \leq 2\|G\|$. Hence, if Ω has finite improved Poincaré constant C_{IP} , then $\|h\|_2^2 \leq 4C_{\text{IP}} \|G\|^2$.*

Proof. By a standard regularization argument, it suffices to prove the claim for smooth h and G ; the general case follows by approximation. We start with the relation between h and G . Using $\nabla h = \operatorname{div} G$ in (i) below,

$$\begin{aligned}\|\delta \nabla h\|_2^2 &= \int_{\Omega} \delta^2 |\nabla h|^2 = \sum_i \int_{\Omega} \delta^2 \partial_i h \partial_i h \stackrel{(i)}{=} \sum_{ij} \int_{\Omega} \delta^2 \partial_i h \partial_j G_{ij} \stackrel{(ii)}{=} - \sum_{ij} \int_{\Omega} \partial_j (\delta^2 \partial_i h) G_{ij} \\ &= -2 \sum_{ij} \int_{\Omega} \delta (\partial_j \delta) (\partial_i h) G_{ij} - \sum_{ij} \int_{\Omega} \delta^2 (\partial_{ij} h) G_{ij} \stackrel{(iii)}{=} -2 \sum_{ij} \int_{\Omega} \delta (\partial_j \delta) (\partial_i h) G_{ij},\end{aligned}$$

where in (ii), we used integration by parts and the fact that $\delta = 0$ on $\partial\Omega$, and in (iii), we used $\langle \nabla^2 h, G \rangle = 0$ since the Hessian of h is symmetric, and G is anti-symmetric. Then,

$$\begin{aligned}\|\delta \nabla h\|_2^2 &= -2 \int_{\Omega} \delta \sum_{i < j} \{(\partial_j \delta)(\partial_i h) - (\partial_i \delta)(\partial_j h)\} G_{ij} \leq 2 \int_{\Omega} \delta \left| \sum_{i < j} \{(\partial_j \delta)(\partial_i h) - (\partial_i \delta)(\partial_j h)\} G_{ij} \right| \\ &\leq 2 \int_{\Omega} \delta \left(\sum_{i < j} \{(\partial_j \delta)(\partial_i h) - (\partial_i \delta)(\partial_j h)\}^2 \right)^{1/2} \left(\sum_{i < j} G_{ij}^2 \right)^{1/2}.\end{aligned}$$

Next, the summation in the first term is equal to $|\nabla \delta|^2 |\nabla h|^2 - \langle \nabla \delta, \nabla h \rangle^2$ and thus at most $|\nabla \delta|^2 |\nabla h|^2 \leq |\nabla h|^2$, since δ is 1-Lipschitz. Therefore,

$$\|\delta \nabla h\|_2^2 \leq 2 \int_{\Omega} \delta |\nabla h| \left(\sum_{i < j} G_{ij}^2 \right)^{1/2} \leq 2 \left(\int_{\Omega} \delta^2 |\nabla h|^2 \right)^{1/2} \left(\int_{\Omega} \sum_{i < j} G_{ij}^2 \right)^{1/2},$$

which completes the proof. $\square$

With these lemmas in hand, we can prove the main theorem of this section.

Proof of Theorem 1.4. Let $f \in Q$. Pick $u \in V$ from Lemma B.2 and G from Lemma B.3 for this u . Then for $h = f - \operatorname{div} u$, we have $\|h\|_2^2 \leq 4C_{\text{IP}} \|G\|^2$ by Lemma B.5. Moreover, using Lemma B.4 and Cauchy–Schwarz,

$$\|G\|^2 = \langle h, \operatorname{div} u \rangle_Q \leq \|h\|_2 \|\operatorname{div} u\|_2 \leq 2\sqrt{C_{\text{IP}}} \|G\| \|\operatorname{div} u\|_2,$$

which implies $\|G\|^2 \leq 4C_{\text{IP}} \|\operatorname{div} u\|_2^2$. Finally,

$$\begin{aligned}\|f\|_2^2 &= \|h\|_2^2 + \|\operatorname{div} u\|_2^2 + 2\langle h, \operatorname{div} u \rangle_Q \leq 4C_{\text{IP}} \|G\|^2 + \|\operatorname{div} u\|_2^2 + 2\|G\|^2 \\ &\leq 4C_{\text{IP}} \|G\|^2 + \|\operatorname{div} u\|_2^2 + \|G\|^2 + 4C_{\text{IP}} \|\operatorname{div} u\|_2^2 = (1 + 4C_{\text{IP}}) (\|\operatorname{div} u\|_2^2 + \|G\|^2) \\ &= (1 + 4C_{\text{IP}}) \|\nabla u\|_2^2 = (1 + 4C_{\text{IP}}) \left(\sup_{v \in V, v \neq 0} \frac{\langle f, \operatorname{div} v \rangle_Q}{\|\nabla v\|_2} \right)^2\end{aligned}$$

which proves the theorem. $\square$